

Deploying an AI Application Using Docker Model Runner

Find out what the recently launched Docker Model Runner is and how to deploy an AI application using it.

Docker has just launched Docker Model Runner to help developers run or test LLM-based applications locally by pulling the model running with the Docker Desktop (minimum version required is 4.40) on localhost. This beta feature must be enabled in the Docker Desktop settings. We can pull different models like ai/smollm2, ai/deepseek-r1-distill-llama, ai/llama3.2, etc, from the Docker model registry at <https://docs.docker.com/ai/model-runner/>, as well as any of the required models on localhost as per our requirements. Once the Docker Model Runner is enabled we can work with the Docker model using the commands shown in Figure 1.

Let's use the ai/smollm2 model in our AI Agent, which is built on top of a Python application that will be using the Model Runner as a backend for all interactive chats. The following code snippet calls the Docker Model Runner:

```
response = requests.post(
    # f"{MODEL_RUNNER_URL}/engines/llama.cpp/v1/
    chat/completions",
```

```
    f"{MODEL_RUNNER_URL}/engines/v1/chat/
    completions",
    json={
        "model": "ai/llama3.2:latest", # Example
        "messages": [{"role": "user", "content":
        model, can be changed
        prompt}],
        "max_tokens": 100
    }
)
```

To deploy this application, we just need to clone the git code (<https://github.com/ashutoshbhakare/dockeraiaagent>) as shown in Figure 2.

**Docker Workshop on LLM Chat
Interface with Docker Model
Runner**

Figure 2: git clone

```
PS C:\Users\cidco> docker model status
Docker Model Runner is running

Status:
llama.cpp: running llama.cpp latest-cpu (sha256:143d9d8f4c2d7c252ec32d7a560869d227c1be39e4ca89c20702e6959a60a39c) version: unknown
PS C:\Users\cidco> docker model
Usage: docker model COMMAND

Docker Model Runner

Commands:
inspect  Display detailed information on one model
list     List the available models that can be run with the Docker Model Runner
logs     Fetch the Docker Model Runner logs
pull     Download a model
push     Upload a model
rm       Remove models downloaded from Docker Hub
run      Run a model with the Docker Model Runner
status   Check if the Docker Model Runner is running
tag      Tag a model
version  Show the Docker Model Runner version

Run 'docker model COMMAND --help' for more information on a command.
PS C:\Users\cidco> docker model pull ai/smollm2
Using cached model: 258.06 MB
PS C:\Users\cidco> docker model list
```

MODEL NAME	PARAMETERS	QUANTIZATION	ARCHITECTURE	MODEL ID	CREATED	SIZE
ai/smollm2	361.82 M	IQ2_XXS/Q4_K_M	llama	354bf30d0aa3	2 months ago	256.35 MiB
ai/gemma3	3.88 B	IQ2_XXS/Q4_K_M	gemma3	0b329b335467	2 months ago	2.31 GiB
ai/llama3.2	3.21 B	IQ2_XXS/Q4_K_M	llama	436bb282b419	2 months ago	1.87 GiB

Figure 1: List of Docker model commands